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The use of ChatGPT as a writing feedback tool in second language learning and teaching: a systematic review

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ABSTRACT

This study systematically reviews 40 key articles (1 January 2023–31 July 2025) on ChatGPT as a writing feedback tool, examining authors' geographic distribution, methodologies, educational settings, language contexts, research foci and gaps. The synthesis identifies four main themes: (a) benefits and limitations of ChatGPT feedback, (b) factors influencing its effectiveness, (c) user experiences and perceptions and (d) pedagogy of ChatGPT-supported teacher or peer feedback and its impact on feedback quality and writing performance. Key gaps include the need for diversified topics, robust theoretical frameworks, exploration of local ChatGPT alternatives, longitudinal studies, revised evaluation criteria, genre-specific impact, application beyond English, research in pre-tertiary settings and ethical considerations. This review offers a comprehensive analysis of ChatGPT's role in writing feedback, highlights future research directions and provides insights for teaching practice.

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KEYWORDS
ChatGPT; artificial intelligence; writing feedback; systematic review; second language learning and teaching

1. Introduction

Feedback is critical for effective second/foreign language writing (Bai and Hu 2017; Zhang, 2022). Providing appropriate information about the language learners' writing performance, feedback is pedagogically a crucial link between assessment and teaching, as well as guidance to achieve the target learning goals. Written Corrective Feedback (WCF) plays a vital role in enhancing learners' writing performance and task achievement (Karim and Nassaji 2020; Leeman 2010; Lim and Renandya 2020; Q. Liu and Brown 2015; Y. Liu and Huang 2020; Luo and Liu 2017; Zhang 2022).

However, providing effective and constructive feedback can be both time-consuming and mentally exhausting (Teng and Ma 2024), often resulting in an overwhelming workload for teachers (McMartin-Miller 2014). In order to liberate teachers from the fatigue of repetitively correcting students' writings (Hyland 1990), researchers are increasingly investigating the potential application of Automated Writing Evaluation (AWE) systems, which involve automatically generating writing improvement suggestions through

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software or tools, offering significant advantages in terms of efficiency and engagement. For example, it has been shown to enhance EFL students' writing engagement (Zhang 2020).

Recently, advances in artificial intelligence have facilitated the integration of AWE systems into second-language writing instruction, transforming students' learning experiences. AWE has been claimed not only to relieve L2 instructors from grading workload (Ranalli 2018) but also to provide students with learning affordances through automated formative feedback (Cotos 2014; Feng, Li, and Zhang 2025; Stevenson and Phakiti 2014).

With the rapid development of AI technology, research interest in the integration of artificial intelligence into second language (L2) writing has grown exponentially (e.g. Fitria 2023; Kwon, Shin, and Lee 2023; Parker, Becker, and Carroca 2023; Su, Lin, and Lai 2023; Warschauer et al. 2023). However, the existing systematic reviews have mainly examined the practical implementation and effectiveness of AWE in foreign language writing classrooms before 2023. For example, Karatay and Karatay (2024) reviewed studies mainly published between 2013 and 2021. There remains a significant gap in studies exploring how AI-driven AWE systems, especially those utilising technologies like ChatGPT, have affected students' revision processes and preferences.

The advent of ChatGPT in late 2022 has been a game-changer, opening new avenues for application in AWE (Mizumoto and Eguchi 2023). While ChatGPT's feedback shares some similarity with traditional automatic writing evaluation, it possesses unique features such as interactive capabilities with human writers. Further exploration into the process of interacting with AI is essential, particularly in areas like crafting and refining prompts to optimise responses.

2. Literature review

2.1. An overview of AWE

Originating from automated essay scoring (AES), AWE systems have been referred to by different terms, including machine scoring of essays, computerised essay scoring and automated essay evaluation (Cotos 2014). AWE systems, employing an array of complex methods, such as machine learning (ML), natural language processing (NLP), artificial intelligence (AI) and advanced statistical methods, help enhance learners' writing competence (Grimes and Warschauer 2010). AWE systems have gained prominence in educational settings due to their pedagogical implication, particularly their provision of immediate feedback on lexical complexity, syntactic variety, discourse structures, lexicogrammar and content development (Z. Li et al. 2014). The integration of AWE systems into language classrooms has been further facilitated by their features of providing prompt feedback, revision opportunities and support for multiple submissions (Godwin-Jones 2018).

2.2. ChatGPT as a feedback tool

ChatGPT demonstrates an additional advantage over traditional AWE tools in providing writing feedback owing to its human-like conversational abilities. ChatGPT is capable of providing personalised and comprehensive feedback on students' writing (Caldarini, Jaf,

and McGarry 2022). As Barrot (2023a, 2023b) noted, ChatGPT's potential benefit in L2 writing instruction lies in its dual role as a rich information source as well as an interactive tool capable of clarifying themes, organising structural components and delivering adaptive feedback in real time. ChatGPT facilitates the feedback process by generating customised prompts or responses that align with the teacher's guidelines and address students' individual needs. The immediate ChatGPT feedback enables learners to comprehend and incorporate corrections and suggestions into their revisions, thereby improving their writing quality.

2.3. Research questions

Despite the growing interest in the use of ChatGPT as a writing feedback tool and repeated calls for synthesising existing research to advance the field, there is a scarcity of review papers that bring together the findings of extant investigations and examine their broader implications. Of these few reviews, B. Li et al. (2024) investigated published research on the ChatGPT application in language learning from November 2022 to November 2023, outlining the types of papers, key topics, methodologies adopted, major research trends and existing gaps demanding attention. In another systematic review of the 20 most pertinent studies on ChatGPT research during 2023 and 2024, Teng (2024c) explored the role of ChatGPT in English as a Foreign Language (EFL) writing. However, this review does not focus specifically on the employment of ChatGPT as a writing feedback tool.

To fill this gap, the current study aims to synthesise research to understand the role of ChatGPT as a feedback tool in second language writing and learners' perceptions of using ChatGPT in writing feedback, guided by the following research questions:

RQ1. How are the current publications on using ChatGPT as a writing feedback tool characterised by geographical distribution of authors, research methodology, educational setting and language context?

RQ2. What are the primary research focus areas in the published articles about using ChatGPT as a writing feedback tool?

RQ3. What are the existing research gaps and potential directions for future research on using ChatGPT as a writing feedback tool?

3. Methodology

The procedure for selecting and screening the literature in this review is based on the PRISMA (Preferred Reporting Items in Systematic Reviews and Meta-analyses) guidelines (Moher et al. 2009) and is shown in Figure 1. Illustrated below are the details about each step taken for the current study.

3.1. Identifying keywords for conducting a literature search

We used terms such as "artificial intelligence" or "AI" or "ChatGPT" or "GPT" or "Chatbot" for AI tool use and "writing feedback", or "writing corrective feedback",

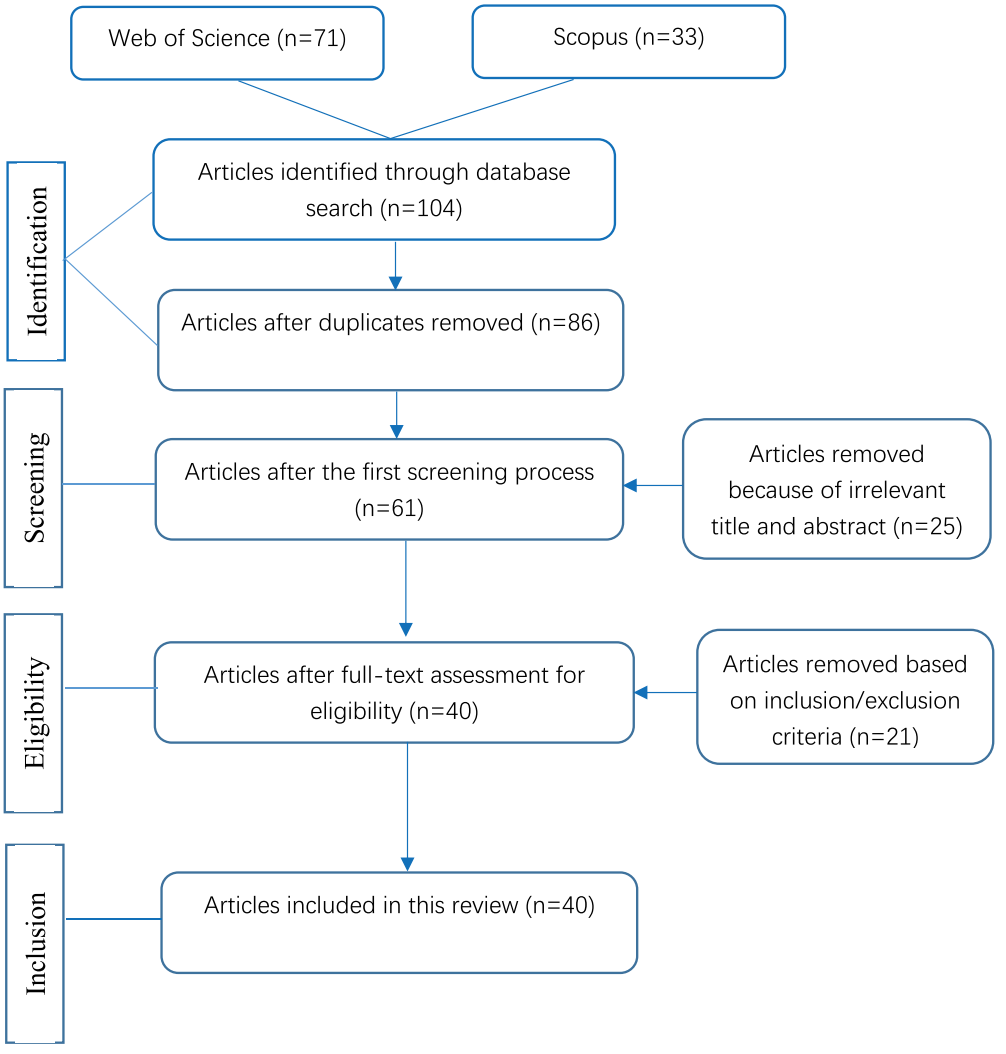


Figure 1. Preferred Reporting Items in Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of study selection.

Table 1. Initial search string.

Topic	Search terms
ChatGPT	"artificial intelligence" or "AI" or "ChatGPT" or "GPT" or "Chatbot"
Writing feedback	"writing feedback", or "writing corrective feedback", or "automatic writing evaluation", or AWE, or "peer feedback" or "teacher feedback".

or "automatic writing evaluation", or AWE or "peer feedback", or "teacher feedback". The search terms are listed in Table 1, which were used to identify the relevant literature from sources.

Table 2. Final inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Published 1 January 2023–31 July 2025	Published before 2023
Published in English	Not published in English
Empirically-based research, including an experiment or case study conducted in second or foreign language contexts	Not empirical research (e.g. reviews or commentaries) and articles related to first language learning
Academic peer-reviewed journal article	Preprint, conference paper, editorial materials, chapters and review articles
ChatGPT feedback, including some alternatives to ChatGPT, particularly conversational tools, e.g. Poe, which allows users to engage in back-and-forth conversations with AI.	Studies on traditional AWE, but not related to the application of ChatGPT in writing feedback
Indexed in Scopus and Web of Science	Papers that are not available in full text

3.2. Literature search on digital libraries

Relevant articles were sourced from the Web of Science and Scopus. These databases were chosen because of their extensive coverage of the research field of language learning, applied linguistics and language assessment. They are highly acknowledged for their comprehensive indexing of reputable academic journals in the field. Using the keywords identified in Step 1, the search for relevant publications was conducted in those databases.

3.3. Criteria for inclusion

The initial keyword search on the two databases yielded 104 articles. Eighteen duplicates were removed from further analysis. Twenty-five additional articles were excluded after the first screening process, as their focus, as revealed from the abstracts, did not align with the scope of this synthesis. The remaining 61 articles went through a further screening process to refine the study scope in accordance with the inclusion and exclusion criteria outlined in Table 2. After applying the inclusion and exclusion criteria, we selected a total of 40 articles for the present systematic review.

To ensure coding consistency for inclusion or exclusion of articles, two experienced language teaching experts, one of whom was also an author, were responsible for article identification and screening from each database. They cooperated with each other during the screening and selection process and resolved disagreements through discussion. The interrater reliability of the percent agreement reached 93%.

4. Results

4.1. Author geographic distributions of selected studies

Figure 2 presents the geographic distribution of authors who have contributed to selected publications on using ChatGPT in providing writing feedback, published between January 2023 and July 2025.

As indicated in Figure 2, China is the leading country with authors producing more articles than those in other countries on this topic, significantly contributing to the research on using ChatGPT as a writing feedback tool. The USA and Saudi Arabia are the second-largest contributors, each with four publications. Authors from various

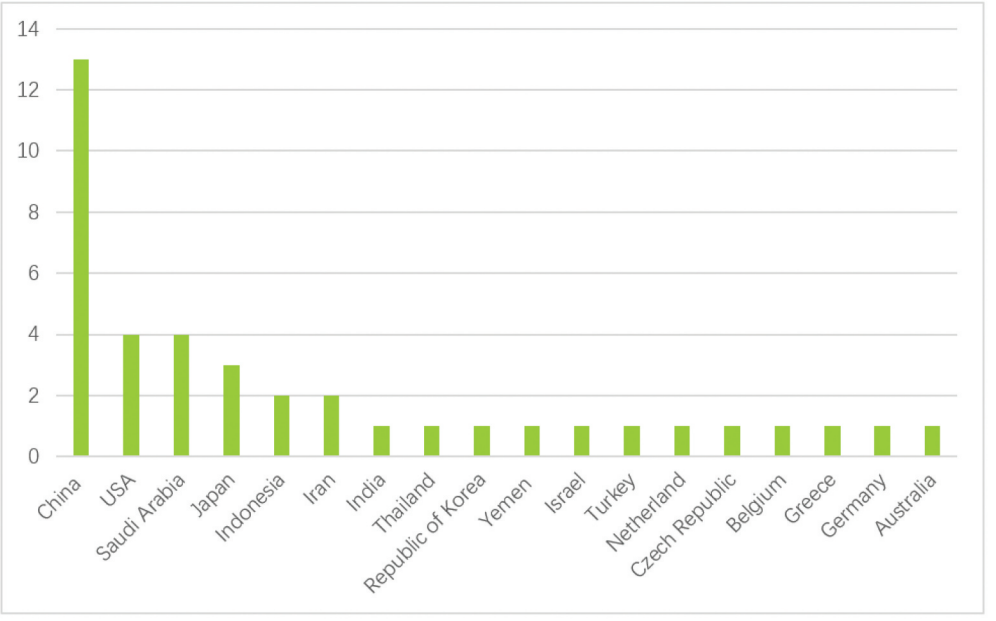


Figure 2. Distribution of selected articles based on their origin of the first author.

countries highlight a noteworthy diversity in the authors' origins, indicating that ChatGPT's role as a writing feedback tool for EFL/ESL students has attracted global attention.

4.2. Theoretical framework adopted in the selected studies

Twenty-nine out of 40 selected studies do not explicitly indicate theoretical orientations. Among the 11 studies confirming their theoretical orientations, four (Alsofyani and Barzanji 2025; Asadi, Ebadi, and Mohammadi 2025; Han and Li 2024; Yan 2024) drew upon sociocultural theory to explore the effects of ChatGPT-generated feedback on EFL learners' writing skills and writing task improvement. For instance, Yan (2024) examined the effects of collaborative processing of ChatGPT-generated feedback on second language writing development. Sociocultural theory can provide robust theoretical support to understand the integration of ChatGPT and traditional teacher feedback in enhancing EFL learners' writing skills. From Vygotsky's perspective, sociocultural theory emphasises the importance of social interaction, cultural tools and mediated learning in cognitive development. AI tools, such as ChatGPT, can be viewed as mediators within the Zone of Proximal Development (ZPD), where learners can lean on to develop skills through guided support (Gauvain 2020; Marginson and Dang 2017). This theoretical lens underscores how technology can serve as a transformative tool to support autonomous learning and teacher–student interactions, addressing the ongoing challenge of developing writing skills proficiency among English language learners (Panofsky 2003). Four studies (Barrios-Beltran 2025; Kurt and Kurt 2024; Mahapatra 2024; Zeevy-Solovey 2024) drew upon multiple theoretical frameworks. For example, Barrios-Beltran (2025) adopted the

framework of Hattie and Timperley's model, the SLA-informed approach and AI in Education together. These three frameworks illustrate the highly interdisciplinary nature of modern language teaching research, particularly when integrated with technology. The study is not confined to traditional branches of linguistics, such as syntax or phonology, but deeply integrates educational psychology, second language acquisition, sociocultural theory and educational technology, serving as a prime example of this interdisciplinary research. Two drew upon second language acquisition, with one following Ellis's theory of error analysis (Al-Garaady and Mahyoob 2023) and the other situated within the complexity, accuracy and fluency (CAF) framework (Mizumoto et al. 2024). One study (Jiang, 2025) drew upon the Interactive, Constructive, Active and Passive (ICAP) theory.

4.3. Method and data source

Demonstrated in Table 3 are the research methodologies employed in the 40 selected empirical articles. Twenty-one out of 40 articles adopted a mixed-method approach, whereas 13 of them took a quantitative approach, and only 6 were based on a qualitative approach. As shown in Figure 3, the selected studies preferred mixed and quantitative approaches, accounting for 52.5% and 32.5%, respectively, while only 15% employed qualitative approaches.

The exploration of data sources (Table 3) reveals that most studies used students' work ($n = 18$), including written texts and samples ($n = 15$), chat logs ($n = 1$), writing e-portfolio ($n = 1$) and classroom discussion records ($n = 1$), followed by interviews ($n = 13$) and questionnaires ($n = 12$). Feedback generated by ChatGPT, teachers or peers were used in 10 studies, tests and assessments in nine studies (e.g. Asadi, Ebadi, and Mohammadi 2025), corpus in 5 studies (e.g. Mizumoto and Eguchi 2023), students' reflection, including reflective journals and self-reflective questions, in three studies (e.g. AlGhamdi 2024), surveys in three studies, observations in two studies (e.g. Han and Li 2024) and classroom or group discussion in two studies (e.g. Mahapatra 2024).

4.4. AI tools utilised in the selected studies

Most AI tools generating feedback in these studies are ChatGPT ($n = 32$). Other AI tools utilised in the selected studies are Poe (an AI-driven application as an alternative to ChatGPT) ($n = 3$), AI chatbot named Eva ($n = 1$), My GPTs' (a customised chatbot builder based on GPT-4) ($n = 1$), Kimi.AI (a large-scale language model optimised for language learning developed by a Chinese company) ($n = 1$), EvaluMate (AI Supported peer review system) ($n = 1$) and Wordtune (an AI-based application) ($n = 1$). As shown in the reviewed papers, various AI technologies have been used in diverse ways to support writing instruction as a complement to teacher's feedback and peers' feedback.

However, most studies regarding the employment of ChatGPT for writing feedback did not specify the model type of ChatGPT. Among a total of 32 studies, only 13 studies specified the model version, with ChatGPT-4 being the most frequently reported in eight studies (Escalante, Pack, and Barrett 2023; Fokides and Peristeraki 2024; Lin and Crosthwaite 2024; Mizumoto et al. 2024; Pfau, Polio, and Xu 2023; Poole and Coss 2024; Shin and Lee 2024; Zou et al. 2025), followed by ChatGPT-3.5 in three studies (Alsofyani and Barzanji 2025; Meyer et al. 2024; Yan 2024) and ChatGPT-3 in two studies (Deygers

Table 3. Key information from sample studies.

The study	Theoretical framework	Method	Data collection method	AI tool	Educational setting	Language context
Alanazi et al. (2025)	unspecified	mixed	writing assessments, Questionnaires and interviews	ChatGPT	university	EFL
Al-Garaady and Mahyoob (2023)	error analysis	mixed	written texts	ChatGPT	university	EFL
AlGhamdi (2024)	unspecified	qualitative	written texts, feedback generated by ChatGPT, and reflections	ChatGPT	university	EFL
Allen and Mizumoto (2024)	unspecified	quantitative	Questionnaire	ChatGPT	university	EFL
Alsofyani and Barzanji (2025)	constructivism and sociocultural theory (ZPD)	mixed	standardized pretest and posttest, survey and open-ended questions	ChatGPT-3.5	university	EFL
Asadi, Ebadi, and Mohammadi (2025)	sociocultural theory (ZPD)	mixed	pre-and post-test assessments, semi-structured interviews	ChatGPT	private language institute	EFL
Banihashem et al. (2024)	unspecified	quantitative	students' essays and feedback generated by peers and ChatGPT	ChatGPT	university	ESL
Barrios-Beltran (2025)	Hattie and Timperley's model SLA-informed approach to the AI in Education framework	qualitative	Questionnaires, Writing Texts, AI-Generated Feedback, Scoring (Instructor & AI), Classroom Discussion Records	ChatGPT-4	university	Spanish as a second language
Bucol and Sangkawong (2024)	unspecified	mixed	essay samples, observation	ChatGPT	university	EFL
Chan, Lo, and Wong (2025)	unspecified	mixed	writing papers, instructor Scoring, Questionnaire, interview	ChatGPT	university	ESL
Deygers et al. (2025)	unspecified	quantitative	Students' written texts, survey	ChatGPT-3	university	EFL
Escalante, Pack, and Barrett (2023)	unspecified	mixed	writing tests, questionnaires, open-ended responses and AI-generated feedback texts	ChatGPT-4	university	ENL (English as a New Language)
Fokides and Peristeraki (2024)	unspecified	mixed	written texts, feedback generated by ChatGPT and teachers	ChatGPT v.4 turbo	primary school	EFL & Greek as a native language
Gozali et al. (2024)	unspecified	qualitative	interview, reflective journals, e-portfolios and peer feedback	ChatGPT	university	EFL
Guo, Pan, et al. (2024)	unspecified	quantitative	written texts, feedback provided by peers	Eva	university	EFL
Guo and Wang (2023)	unspecified	mixed	written texts, feedback generated by ChatGPT and teachers, questionnaires	ChatGPT	university	EFL
Guo, Zhang, et al. (2024)	unspecified	mixed	chat log	EvaluMate	university	EFL
Han and Li (2024)	sociocultural theory, Mediation scaffolding	mixed	written texts, ChatGPT-supported teacher feedback, class observations and interviews	ChatGPT	university	EFL
Jamshed et al. (2024)	unspecified	quantitative	pretest and posttests, Questionnaires	ChatGPT	secondary	ESL

(*Continued*)

Table 3. (Continued).

The study	Theoretical framework	Method	Data collection method	AI tool	Educational setting	Language context
Jiang (2025)	The Interactive, Constructive, Active, Passive (ICAP)	mixed	pretest and posttests, semi-structured interview	Kimi.AI	university	CSL
Kurt and Kurt (2024)	Feedback thinking from a socio-constructivist approach AI literacy	qualitative	interviews	ChatGPT	university	EFL
Lin and Crosthwaite (2024)	unspecified	mixed	WCF was provided by both teachers and ChatGPT	ChatGPT-4	high school or university	mixed
Y. Liu, Lu, and Qi (2025)	unspecified	quantitative	written essays from the TOEFL11 corpus	ChatGPT	unspecified	EFL
Mahapatra (2024)	Feedback as a dialogic process Barrot's theory of ChatGPT as a reliable writing tool	mixed	tests of writing, focus group discussions	ChatGPT	university	ESL
Meyer et al. (2024)	unspecified	quantitative	students' written texts, scale, questionnaires	GPT-3.5-turbo	upper secondary	EFL
Mizumoto and Eguchi (2023)	unspecified	quantitative	Corpus of TOEFL11	ChatGPT-3.5 - davinci-003 model	unspecified	EFL
Mizumoto et al. (2024)	complexity, accuracy and fluency (CAF)	quantitative	Cambridge Learner Corpus First Certificate in English	ChatGPT-4	unspecified	EFL
Oktarin et al. (2024)	unspecified	mixed	English writing test, questionnaire and interviews	ChatGPT	university	EFL
Pfau, Polio, and Xu (2023)	unspecified	quantitative	corpus of Greek learners of English	ChatGPT-4	unspecified	EFL
Polakova and Ivenz (2024)	unspecified	mixed	pretests, post-tests,8 questionnaires and focus group interviews	ChatGPT	university	EFL
Poole and Coss (2024)	unspecified	quantitative	a corpus of third-semester university L2 Chinese learners	ChatGPT-4	university	EFL
Rad, Alipour, and Jafarpour (2024)	unspecified	mixed	writing tasks, Scale, Semi-structured interview	Wordtune	private language institute	CSL
Shin and Lee (2024)	unspecified	quantitative	Written essays	My GPTs	high school	EFL
Tam (2024)	unspecified	qualitative	self-reflective questions, interviews and conversation logs	ChatGPT	university	ESL
Teng (2024a)	unspecified	mixed	Questionnaire, interviews	Poe	university	EFL
Teng (2024b)	unspecified	mixed	Questionnaire, interviews	Poe	university	EFL
Wang (2024)	unspecified	quantitative	Tests and a writing anxiety scale	Poe	university	EFL
Yan (2024)	sociocultural theory (SCT)	mixed	Written Texts (Drafts and Final Products)	ChatGPT-3.5	university	EFL
Zeevy-Solovey (2024)	strategy of collaborative writing, process approaches and multiple revisions	qualitative	Checklist, feedback and survey	ChatGPT	university	EFL
Zou et al. (2025)	unspecified	mixed	Written texts, Questionnaire, Feedback comments	ChatGPT-4	university	EFL

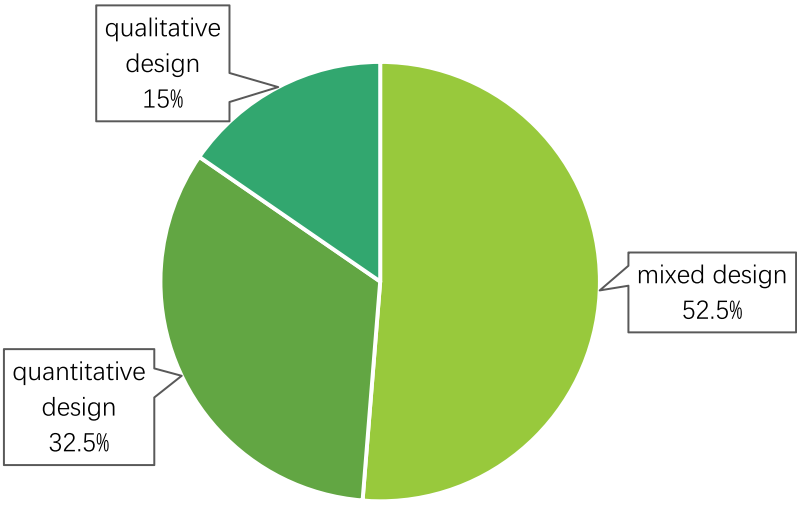


Figure 3. Methodologies adopted in the selected studies.

et al. 2025; Mizumoto and Eguchi 2023). As suggested in the research conducted by Y. Liu, Lu, and Qi (2025), such technical factors as GPT model type, fine-tuning methods, and dataset characteristics played a critical role in the effectiveness of ChatGPT for writing feedback. It is assumed that the performance of GPT's later versions would surpass that of its predecessor, with the potential to generate more sophisticated language processing and human-like responses (Rahaman et al. 2023). For instance, Y. Liu, Lu, and Qi (2025) revealed that ChatGPT 4 models outperformed ChatGPT 3.5 models in their tests. Fokides and Peristeraki (2024) also employed ChatGPT 4 instead of ChatGPT 3.5 due to ChatGPT 3.5's deficiency in accurately correcting texts written in Greek. Therefore, it is advisable that studies exploring the effectiveness of ChatGPT should report the GPT model type and other relevant information so that there will be more transparency regarding the AI tool generating the writing feedback. Since ChatGPT is constantly being updated and refined, researchers should always note the model type and version of ChatGPT and take the technical factors into consideration when analysing the effectiveness of ChatGPT.

4.5. Educational setting and language context

Findings show that higher education was the dominant setting and English was the primary target language. A majority of the studies targeted a tertiary-level context to investigate ChatGPT's use in providing writing feedback, with three studies at the high school or secondary level and only one study at the primary level, indicating a scarcity of studies in the educational setting below the tertiary level. In terms of the language of AI (ChatGPT) systems, all selected articles investigated an AI (ChatGPT) system that included feedback in English, except one in Chinese (Jiang 2025). Most of the studies targeted EFL settings ($n = 30$), followed by ESL ($n = 5$), mixed (ESL and EFL) ($n = 1$), ENL ($n = 1$), Spanish as a second language ($n = 1$) and Chinese as a second language ($n = 2$). While most reviewed articles focused on EFL or ESL, an emerging trend has begun to extend this line of research to the application of AI in learning languages other than English.

4.6. Topic foci (research focus areas)

Four research focus areas for using ChatGPT as a writing feedback tool were noted in the following analysis.

4.6.1. The benefits and constraints of ChatGPT feedback

Offering significant benefits in efficiency, accuracy, reliability and consistency, ChatGPT represents an innovative tool for writing feedback, though existing evidence is largely derived from short-term, small-scale studies with limited contextual diversity.

One of ChatGPT's most significant advantages for EFL or ESL learners is the efficiency and accessibility of feedback. Unlike traditional teacher or peer feedback, which often involves time delays, ChatGPT, provides instant responses on grammar, sentence structure and content organisation (Alanazi et al. 2025).

In addition, Mizumoto and Eguchi (2023) indicated the accuracy and reliability of automated corrective feedback using GPT technology, suggesting that GPT-based systems could be effectively integrated into classroom teaching to assist with essay scoring, providing students with timely feedback to improve their writing skills and reducing the workload on instructors (Mahapatra 2024).

A body of studies (Z. Jiang et al. 2023; Mizumoto et al. 2024; Pfau, Polio, and Xu 2023) has examined ChatGPT's viability for measuring linguistic accuracy in second language writing, highlighting ChatGPT's capability to provide reliable assessments of writing accuracy. For instance, Z. Jiang et al. (2023) noted that LLMs, particularly ChatGPT-4, exhibited great potential in assessing L2 writing accuracy with high precision. According to Bucol and Sangkawong (2024), ChatGPT could accurately assess students' writing using prompts and pre-defined rubrics, outperforming human raters in effectiveness, speed, consistency and scalability. Its advent has introduced a new paradigm in providing writing feedback, demonstrating great potential for both writing improvement and assessment. Nevertheless, most studies focus on accuracy rather than holistic writing quality, and few adopt blind or double-blind designs, raising concerns about potential assessment bias.

Furthermore, findings by Meyer et al. (2024) highlighted that using ChatGPT could yield cognitive and affective-motivational benefits by providing individualised instruction to all students. Chan, Lo, and Wong (2025) also reported that students receiving AI feedback demonstrated higher levels of motivation and engagement, apart from their improvements in writing quality during the revision process. Notably, several key limitations of these studies are also acknowledged, including their short-term task design, variability in AI-generated feedback quality and lack of causal evidence. Motivational outcomes are often measured via self-report questionnaires, which may be subject to response bias. The sustained motivational effects have not yet been systematically documented, and longer-term impacts on student writing skills need to be investigated.

However, the limitations of ChatGPT used as a writing feedback tool were also underscored in some studies. For example, Al-Garaady and Mahyoob (2023) revealed that ChatGPT successfully identified most surface-level errors but failed to detect writing errors related to deep structure and pragmatics, which human instructors were able to identify. This suggests that while ChatGPT can serve as a valuable tool for enhancing L2 writing

accuracy, it cannot fully substitute for human raters. The evidence indicates that ChatGPT functions best as a complementary tool alongside human instruction. S. Li (2025) claimed GenAI ratings were more consistent with human ratings when GenAI was trained using a large number of scored essays and when the rating criteria were well-defined, thereby suggesting a need for criteria-based feedback and feedback evaluation. Deygers et al. (2025) maintained that while ChatGPT exerted a significant impact on text length and complexity, it did not necessarily facilitate sustained improvement in writing quality. More studies in the future will be needed to evaluate the long-term effect of ChatGPT's feedback on language learning outcomes.

4.6.2. Students' and teachers' experiences and perceptions in using ChatGPT as a writing feedback tool

EFL students have demonstrated a strong preference for using ChatGPT in their writing practices due to its effective feedback and user-friendly interface. For example, a study conducted by Allen and Mizumoto (2024) indicated that Japanese EFL students highly favoured ChatGPT, driven by its ability to provide effective feedback, significantly enhancing the clarity and cohesion of their writing. Teng (2024) explored the EFL learners' perceptions and experiences in using ChatGPT for generating writing feedback. The finding reported that students developed more confidence in their writing abilities and felt more motivated to write when using ChatGPT. Additionally, ChatGPT exerted positive effects on learners' self-efficacy and collaborative writing tendencies, fostering deeper engagement with writing tasks and encouraging collaborative efforts. Mahapatra (2024) also found that ChatGPT had a favourable impact on students' academic writing skills, with students' perceptions of this impact being overwhelmingly positive as well. The study by Zeevy-Solovey (2024) indicated that both ChatGPT and teacher feedback were perceived as effective, with peer feedback also regarded as effective by most participants.

Overall, the aforementioned studies (Allen and Mizumoto 2024; Teng 2024a; Mahapatra 2024; Zeevy-Solovey 2024) collectively highlight the positive impact of ChatGPT on EFL students' writing performance and attitudes. By enhancing clarity, cohesion, motivation and engagement, ChatGPT emerges as a valuable tool in EFL writing, improving both individual and collaborative writing experiences.

While most researchers advocated the positive impact of ChatGPT on EFL students' writing performance and attitudes, some others observed different patterns in students' perceptions of ChatGPT. The study by Zou et al. (2025) revealed that students displayed a particular inclination towards ChatGPT in addressing organisational issues while recognising teacher feedback in relation to language and content. Kurt and Kurt (2024) explored Turkish pre-service teachers' perceptions of and preferences for the effectiveness of ChatGPT feedback compared with teacher feedback and peer feedback. Their study indicated that participants perceived ChatGPT as having certain constraints, such as occasional inconsistencies and reliance on the quality of prompts despite its practicality, interactivity and adaptability. Barrios-Beltran (2025) reported that while students appreciated the efficiency and structured nature of AI-generated guidance, they simultaneously placed high value on the personalised, contextually sensitive feedback provided by their instructor. At the same time, students expressed ethical concerns – such as overreliance, data privacy, unclear institutional policies and plagiarism risks. These insights underscore

the need to incorporate AI literacy into education to promote critical, ethical and effective use of such tools.

The perception of ChatGPT's feedback has also been explored from the perspective of EFL teachers. As collaborators with ChatGPT, teachers' perceptions of ChatGPT feedback are also of great importance, given that they will not only influence how appropriately AWE technology is integrated into classroom instruction but also provide a key source of evidence for evaluating its efficacy. According to Guo and Wang (2023), on the one hand, the teachers perceived ChatGPT feedback as valuable due to its feedback features, such as the provision of appropriate revision suggestions as well as the rationale behind those suggestions, frequent use of compliments for student achievements and flexible choice of language when giving comments. On the other hand, EFL teachers also reported ChatGPT's drawbacks in its feedback features, with respect to both content and accessibility. The feedback was often described as lengthy, difficult for low-proficiency students to comprehend and sometimes irrelevant or deficient in identifying specific writing problems. Additionally, teachers noted that the feedback could be hard to locate.

4.6.3. Factors influencing the effectiveness of ChatGPT feedback

Many researchers have conducted investigations into the factors influencing the effectiveness of ChatGPT as a writing feedback tool. Y. Liu, Lu, and Qi (2025) have mainly explored potential factors that influence ChatGPT's performance, such as fine-tuning methods, GPT model type (GPT-3.5 vs. GPT-4), prompting strategy (zero-shot vs. few-shot) and dataset characteristics. Their findings indicated that fine-tuned GPT models consistently outperformed non-fine-tuned GPT-4 models, which in turn outperformed GPT-3.5 models. Few-shot prompting did not exhibit clear advantages over zero-shot prompting in their study. It is also revealed that class imbalance and dataset shift negatively affected model accuracy and reliability. These results suggested that such factors as fine-tuning methods, GPT model type, prompting strategy and dataset characteristics were key determinants of the effectiveness of GPT models in writing feedback provision, offering valuable insights into the variables that could influence ChatGPT's performance in AWE. Fokides and Peristeraki's findings (2024) revealed that ChatGPT demonstrated better performance in English short essays than in Greek ones in both the quantitative and qualitative parameters that were examined, highlighting areas for improvement in its application to languages other than English.

The performance of ChatGPT as a writing feedback tool is not only determined by its own capability but also by other factors, such as writers' proficiency levels, metacognitive awareness and using strategy also play an important role in its effectiveness. It was observed that the performance of ChatGPT's assessments and human evaluations diminished with lower proficiency levels (Pfau, Polio, and Xu 2023). Its effectiveness may be more pronounced among intermediate and advanced learners, as its performance appears to correlate with the writers' proficiency levels. Teng's findings (2024b) further revealed that students with higher metacognitive awareness in writing held a statistically significant advantage over those with lower awareness in perceiving ChatGPT's effectiveness as a writing feedback tool. Moreover, the study by Kurt and Kurt (2024) indicated that the effectiveness of ChatGPT depended heavily on whether its users could input clear instructions, emphasising the significance of skilful prompting. Similarly, Fokides and Peristeraki (2024) asserted that it was imperative to craft a thorough prompt, as the

effectiveness of ChatGPT in executing any task hinges on the clarity of the task's outline and the specificity of the instructions given. Poole and Coss (2024) also argued that the accuracy and reliability of ChatGPT's employment as a writing feedback tool can be increased with well-crafted prompts. They claimed that ChatGPT could be a valuable writing assessment tool, provided that it was used strategically with effective prompting strategies.

To sum up, the effectiveness of ChatGPT's feedback was influenced by various factors, such as the utilisation of the system's multilingual capabilities, learners' proficiency level, metacognitive awareness, personal goal and their self-regulated writing strategies, which in turn affect the quality of prompts (Tam 2024). This can explain why previous studies have reached varying conclusions regarding the accuracy and effectiveness of ChatGPT as a writing feedback tool. Meanwhile, it also inspires teachers to take into account the diversity and complexity of influencing factors when designing writing instruction in a second language writing classroom, to develop teaching plans reasonably. For instance, the proficiency levels of their students should be taken into consideration in order to effectively utilise ChatGPT as a writing feedback tool. To ensure key goals are met at every stage of the writing process, instructors should adapt their strategies to address the specific needs of individual learners. Teachers can refine their prompting strategies, such as incorporating more details into queries to elicit relevant answers, instructing ChatGPT to assume a specific role, using delimiters to mark distinct sections of a prompt, outlining steps required to accomplish a task and prompting the model to reflect on those steps, providing illustrative examples and specifying the desired length of the output (Poole and Coss 2024).

4.6.4. The pedagogy of ChatGPT-supported teacher feedback or peer feedback, as well as its impact on students' feedback quality and writing performance

Despite ChatGPT's advantages in identifying surface-level errors and offering reliable automated feedback, its limitations in detecting pragmatic errors (Al-Garaady and Mahyoob 2023) and its inconsistent effectiveness across different writing registers necessitate a balanced approach that combines the use of ChatGPT with human instruction (Teng, 2024c). With a thoughtful, teacher-supervised implementation, ChatGPT's benefits can be maximised while mitigating its drawbacks, ultimately enhancing both teaching and learning experiences in EFL writing settings. The synthesis of the study suggests that while ChatGPT can function as a valuable preliminary tool for error detection, it needs to be complemented by human supervision for more subtle writing issues (Al-Garaady and Mahyoob 2023).

Several recent studies have compared teacher feedback with ChatGPT-generated feedback and highlighted the prospect of teacher-AI collaboration by leveraging ChatGPT to deliver writing feedback. For instance, Guo and Wang (2023) investigated ChatGPT's potential to support teacher feedback in EFL writing. Their study revealed that ChatGPT generated significantly more feedback than teachers, effectively addressing content, organisation and language-related issues. This finding emphasises ChatGPT's capability of enhancing the feedback process and suggests that teachers use ChatGPT to support their own feedback. Another study by Zou et al. (2025) investigates how students incorporate feedback received from both teachers and ChatGPT into their essay revisions. These findings collectively highlight the importance of a collaborative approach, whereby

teachers systematically integrate and supervise the use of ChatGPT to ensure that learners can benefit from ChatGPT's capacities while receiving support and guidance from their instructors at the same time.

Furthermore, Han and Li (2024) explore an innovative "AI + Teacher" model of writing pedagogy by integrating ChatGPT into teachers' provision of writing feedback to alleviate the challenges of providing feedback in a large class. They found that teacher feedback supported by ChatGPT addressed various error categories and included valuable comments on the overall rhetoric. Furthermore, the students incorporated more of the feedback into their revisions across tasks, reflecting their deeper engagement with the feedback content.

Other studies have focused on ChatGPT-supported peers' feedback as well as its impact on students' feedback quality and writing performance. Guo, Pan, et al. (2024) and Guo, Zhang, et al. (2024) investigated how an AI-supported peer feedback approach affected the university EFL students' feedback quality. The results of their studies revealed that incorporating AI supervision enhanced the quality of students' peer reviews and thereby improved their feedback literacy.

In the approach designed by Guo, Pan, et al. (2024), a student reviewer first evaluates a peer's essay and then an AI tool assesses the student reviewer's comments, providing suggestions to improve the quality of the student reviewer's feedback. The student reviewer subsequently revises their initial comments based on AI-generated feedback. This intervention significantly enhanced students' feedback quality. This research also indicated that the proposed approach improved the writing ability of feedback providers, underscoring the potential of AI technology in enhancing peer feedback's quality.

Guo, Zhang, et al. (2024) investigate whether using an AI-supported peer review system (EvaluMate) can enhance the quality of L2 students' peer review comments. The results of their study revealed that the incorporation of AI supervision improved the quality of students' peer reviews. Students adopted diverse strategies – such as introducing new points, adding details and providing illustrative examples, which contributed to enhancing their comment quality.

5. Discussion

The research gaps and future directions identified in this systematic review underscore the need for further investigations to fully understand the use of ChatGPT as a writing feedback tool.

5.1. The need for more diversified research topics

Regarding research topics, the frequent comparison of human feedback with ChatGPT feedback and students' or EFL learners' perceptions of ChatGPT feedback were the primary focus, whereas only limited research centred on teachers' attitudes towards ChatGPT feedback, teaching practices and feedback literacy development. This imbalance suggests a need for more studies that include teachers to directly measure their teaching practices and provide feedback on literacy development from using ChatGPT and to assess their attitudes towards ChatGPT feedback.

In addition, learners' classroom behaviours regarding ChatGPT use for writing feedback provision received scarce attention. To address this gap, future research could adopt in-depth qualitative methods, such as multimodal conversation analysis using video recordings, to closely monitor human-AI interaction in the process of students using ChatGPT to provide writing feedback.

Other research gaps include the need for future researchers to investigate the impact of ChatGPT-supported writing feedback on various specific genres and micro-aspects of writing (Mahapatra 2024).

5.2. The need for diverse research designs

ChatGPT's application in providing writing feedback requires research in diverse contexts to validate its pedagogical benefits in actual classrooms. A majority of selected studies focused on higher education and used college students as their research subjects, possibly due to the convenience of data collection, as most researchers are often university faculty members. This limits the understanding of the effectiveness of ChatGPT feedback in environments typical for young learners. Thus, future studies should be extended to the primary and secondary settings for a comprehensive understanding across all educational levels.

Most of the selected studies have targeted EFL or ESL settings, centring on English, which restricts our understanding of the effectiveness of ChatGPT feedback across multilingual contexts and underrepresented languages (Dai, Suzuki, and Chen 2025; Lee, Jeon, and Choe 2025). The English-centric approach limits knowledge about how GenAI addresses languages with different orthographic and grammatical systems and cultural representations, potentially neglecting important lingual-cultural aspects of language learning (Crompton et al. 2024). While most reviewed articles focused on EFL or ESL, future research should address these limitations by extending this line of research to the application of ChatGPT for writing feedback provision in languages other than English.

This review underscores the need for more diverse research approaches. While selected studies demonstrated a variety of approaches, including quantitative, qualitative and mixed-methods, there remains a scarcity of qualitative methods. Data collection has predominantly relied on students' work, interviews and questionnaires. Moreover, qualitative investigations mainly used self-reported data, such as interviews, reflections and essays, to examine students' perceptions and writing revisions. Therefore, future research should expand the data sources to capture students' dynamic interactions with ChatGPT across various language learning contexts, including observational notes documenting real-time classroom behaviours, video recordings for fine-grained analysis of learner–technology interactions, transcripts of group discussions to illuminate collaborative processes, reflective journals that shed light on learner autonomy, and multimodal artefacts like digital portfolios derived from in-class activities (Du and Alm 2024; Ji, Han, and Ko 2023; Van Horn 2024).

Moreover, longitudinal studies are needed to evaluate the sustained effect of ChatGPT's feedback on language learning outcomes. Such research would enable a comprehensive assessment of the long-term viability and efficacy of ChatGPT-supported writing feedback, thereby guiding the development of long-term strategies for its pedagogical application. However, most existing research has only yielded valuable

insights into the immediate effects of ChatGPT's feedback on learners' writing performance and engagement (Mahapatra 2024; Teng 2024a). Deygers et al. (2025) attempted to examine the longitudinal gains in the writing performance of students who used ChatGPT as a writing evaluation tool. Nevertheless, as acknowledged by the authors themselves, the intervention period, despite being longer than most other studies in this field, was actually insufficient to capture long-term writing gains. Therefore, there is a critical need to investigate the long-term impact of ChatGPT feedback on students' writing performance over extended periods (Alanazi et al. 2025).

Finally, one of the most pressing research gaps is the exploration of locally produced alternatives to ChatGPT in different countries, e.g. Doubao or DeepSeek in China, and the conduct of a comparison between these locally produced AI tools and ChatGPT. The employment of diverse GenAI tools is needed to fully harness their capabilities to provide writing feedback for students. As ChatGPT just represents only one possible form of AI-supported writing feedback, future studies may investigate more diverse types of AI tools to evaluate their performance. Given that ChatGPT is not accessible in all countries, its internationally available substitutes (e.g. metaAI and Claude) or locally produced options (such as Bytedance and Baichuan AI in China) could also be explored for evaluating its role in the future research (Lin and Crosthwaite 2024).

5.3. The need for a well-established theoretical framework

Only 11 out of the 40 selected studies (27.5%) specify their theoretical orientations, which are social cultural theory, second language acquisition, educational psychology (ICAP) and higher education technology. However, no selected studies have adopted complex dynamic systems as a theoretical framework. Current research indicates that EFL learners' writing development exhibits nonlinear characteristics, along with distinct individual differences and dynamism. This suggests that second language writing development possesses typical features of a dynamic complex system. Therefore, future research could adopt the perspective of complex dynamic systems theory to conduct longitudinal studies on the impact of ChatGPT feedback or ChatGPT-supported teachers' feedback on students' development of writing skills, critical thinking, feedback literacy and engagement. The perspective of complex dynamic systems theory will allow researchers to gain a deeper understanding of the complex processes of ChatGPT-supported feedback provision and is essential for informing the development and implementation of AI technologies in ChatGPT-supported feedback activities.

The other 29 studies (72.5%) do not explicitly indicate their theoretical orientations, closely aligning with Riazi, Shi, and Haggerty's (2018) finding in JSLW studies. Riazi, Shi, and Haggerty (2018) revealed that among the top 10 research foci in *the Journal of Second Language Writing* (1992–2016), the categories of feedback (30/47, 63.8%) and assessment (15/20, 75.0%) contained the largest proportions of articles not positively identified with a theoretical orientation. One possible reason is that many studies on the use of ChatGPT as a writing feedback tool are classroom teaching practice-oriented rather than driven by specific theories. The rapid technological innovation also contributes to the large proportions of articles not explicitly identified with a theoretical orientation. Technological advancements often occur faster than the evolution of learning theories. Despite the rapid pace of AI innovations, some educational theories may not have evolved sufficiently

to address the nuances of learning and teaching in an AI era. Traditional theories might not fully account for how AI tools can transform second language learning and teaching practice.

With ChatGPT increasingly being integrated into the process of writing feedback provision, it is essential to rely on a robust theoretical framework to guide both design and practice so as to ensure the sustainability and profound impact of its educational applications. For example, Allen and Mizumoto (2024) indicated in their study that a theoretical basis for developing the questionnaire items was needed. Future research should address this limitation by grounding the questionnaire development process in a well-established theoretical framework.

While tech developers rapidly introduce new tools and capabilities, educational theorists and practitioners may take longer to understand, evaluate and integrate these innovations into established pedagogical frameworks. As AI technologies like ChatGPT continue to advance, there is a need for educational theories to evolve and account for the unique affordances and challenges of these tools in learning contexts.

5.4. Redefining assessment paradigms in the context of widespread use of ChatGPT

There is a pressing need for in-depth research into adjusting or redefining the evaluation criteria for the students' compositions in the context of widespread use of ChatGPT. Traditional assessment frameworks, which emphasise linguistic structure and content, might not be well founded, as the composition may be polished or even entirely generated by ChatGPT. If students are allowed to use ChatGPT as a feedback tool, how teachers should evaluate their compositions deserves great attention. One pertinent consideration is whether students' ability to effectively utilise ChatGPT should be included as an evaluation criterion. For example, assessment could take into account how students formulate prompts to obtain useful feedback from ChatGPT, and whether they are able to critically analyse and apply that feedback to improve their writing. In light of this, teaching practices might require students to submit their essays along with complete chat records from their interactions with AI, including the prompts sent to the AI and the revisions made after receiving AI-generated feedback. Teachers could then provide comprehensive evaluations that encompass students' writing proficiency, AI literacy (see, e.g. Xie, Zhang, and Wilson 2026) and critical thinking skills. Consequently, the widespread use of ChatGPT calls for the incorporation of additional assessment dimensions.

5.5. The need for ethical considerations about the use of ChatGPT

Finally, ethical considerations pertaining to the use of ChatGPT in language education deserve further attention (Alanazi et al. 2025). Future research should examine critical concerns, such as academic integrity – specifically exploring strategies to mitigate over-reliance on inaccurate information generated by AI and curb the temptation for students to commit plagiarism (B. Li et al. 2024). It is imperative for researchers to establish robust ethical frameworks and guidelines to govern the responsible integration of ChatGPT in educational settings, ensuring its application upholds learners' rights, fosters equity and inclusion and preserves the integrity of the pedagogical process.

6. Conclusion and implications

The findings presented in this systematic review of Using ChatGPT as a writing feedback tool between 1 January 2023, and 31 July 2025, reveal key information about selected studies, research focus areas, research gaps and future directions in the field. Firstly, ChatGPT's role as a writing feedback tool for EFL/ESL students has attracted global attention with a noteworthy diversity in the authors' origins. China is the leading country contributing most articles on this topic, followed by the USA and Saudi Arabia. Second, empirical research articles preferred mixed-methods and quantitative approaches, accounting for 52.5% and 32.5%, respectively, while only 16% were qualitative studies. Third, higher education receives the most research attention, with only three of the articles targeting the context of high school or secondary level. Most of the studies targeted EFL settings, followed by ESL, mixed (ESL and/or EFL), ENL, Spanish as a second language and Chinese as a second language. Fourth, the effectiveness and benefits of ChatGPT as a writing assessment tool are central research themes. Students' experiences and perceptions in using ChatGPT as a writing feedback tool are also gaining attention. Fifth, the pedagogy of the ChatGPT-supported teacher's feedback or peer's feedback and its impact on students' feedback literacy are key areas of inquiry.

The research identified several gaps, including the need to 1) measure teachers' teaching practices and feedback literacy development from using ChatGPT and to assess their attitudes towards ChatGPT feedback, 2) adopt the perspective of complex dynamic systems theory to gain a deeper understanding of the complex processes of ChatGPT-supported feedback provision, 3) explore locally produced alternatives of ChatGPT in different countries and conduct a comparison between these locally produced AI tools and ChatGPT, 4) conduct longitudinal studies, 5) adjust or redefine the evaluation criteria for the students' compositions in the context of widespread use of ChatGPT, 6) apply ChatGPT in writing feedback other than English and conduct studies under the educational settings below tertiary level, 7) investigate the impact of ChatGPT-supported writing feedback on various specific genres and micro aspect of writing and 8) examine ethical concerns.

Overall, the exploration of diverse themes highlights the multifaceted role of ChatGPT in providing writing feedback. ChatGPT demonstrates significant potential as a writing feedback tool. Its ability to deliver timely and effective feedback, support precise error detection and offer valuable insights into targeted writing improvement makes it a valuable asset to language learning.

However, given the relatively short-time frame covered in this synthesis, the current state of research topics during this period is confined to a limited body of empirical research findings. While we acknowledge that selected articles may constrain the depth of empirical evidence examined, more diverse and impactful studies shall be anticipated in the coming years as the field continues to mature.

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